

22683

12425

3 Hours / 70 Marks

Seat No.

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- Instructions :**
- (1) All Questions are *compulsory*.
 - (2) Answer each next main Question on a new page.
 - (3) Illustrate your answers with neat sketches wherever necessary.
 - (4) Assume suitable data, if necessary.

Marks

1. Attempt any FIVE of the following :

10

- (a) Define :
 - (i) Feature scaling
 - (ii) Feature selection
- (b) Enlist application of Random forest algorithm.
- (c) Define KNN algorithm.
- (d) Enlist advantages & disadvantages of Baye's theorem.
- (e) Enlist four causes for failure of k-mean algorithm.
- (f) Define Artificial Neural Network with example.
- (g) Define :
 - (i) RNN
 - (ii) Fine tuning



- 2. Attempt any THREE of the following : 12**
- (a) Explain working of Naive Baye's theorem.
 - (b) Explain how does k-mean algorithm work in detail with steps.
 - (c) Explain why we use Random forest ?
 - (d) Explain learn Hyper parameter basis.
- 3. Attempt any THREE of the following : 12**
- (a) Compare feature scaling & feature selection.
 - (b) Write advantages & disadvantages of KNN algorithm.
 - (c) Describe Dimensionality Reduction.
 - (d) Write steps for image classification using convolutional neural network.
- 4. Attempt any THREE of the following : 12**
- (a) Write advantages & disadvantages of decision tree.
 - (b) Explain support vector machine with suitable example.
 - (c) Explain feed forward & back propagation.
 - (d) Describe Gated Recurrent Unit (GRU)
- 5. Attempt any TWO of the following : 12**
- (a) Explain Greedy layerwise pre-training with suitable example.
 - (b) Write steps for working of Random forest algorithm with example.
 - (c) Write a program in python to implement unsupervised machine learning k-mean algorithm.
- 6. Attempt any TWO of the following : 12**
- (a) Describe ANN concepts.
 - (b) Write step by step working of KNN algorithm with example.
 - (c) Write a python program to implement decision tree for classification using suitable data/dataset.
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